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Multi-axis mixing

Abstract

A multi-axis mixing apparatus is disclosed that includes a housing and a controller operatively connected to a first motor and a second motor. The apparatus includes a frame supporting the first motor configured to selectively rotate a rotatable assembly. The rotatable assembly supports and provides an operative connection to the second motor configured to selectively rotate a rotatable sub-assembly. The rotatable sub-assembly includes a mechanism configured to receive and hold a container containing contents to be mixed, and the controller is configured to independently operate the first and second motors to mix the contents.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This application claims the benefit of U.S. Provisional Patent Application No. 63/055,434, filed Jul. 23, 2020, the entire contents of which are incorporated herein by reference in their entirety.

BACKGROUND
(1) The present invention is directed to mixing, and is more specifically directed to multi-axis mixing, related methods, and features thereof.
(2) It is frequently desirable to mix various substances. For example, generally liquid and/or semi-liquid with solids therein such as paint frequently start as various separate substances or colors that are then mixed into a desired mixture, such as to achieve a certain color, tone, hue, reflectivity, and the like. In order to promote more complete and/or more even mixing, various mixers can utilize more than one independent movement during mixing.

(3) At present, mixers are complex. Often, mixers include undesirable components that can require frequent maintenance and/or repairs. Furthermore, containers with contents to be mixed are frequently heavy and/or bulky, and can be difficult to insert and/or remove from the mixing apparatus.
(4) Therefore, there is a need for an improved mixing apparatus and related methods that provides a simplified, more reliable, and easier-to-use mixing and related container handling experience.
SUMMARY
(5) The present invention overcomes shortcomings of the prior art by introducing multi-axis mixing with improved mixing, extraction, electrical, and mechanical properties.
(6) According to a first embodiment of the present disclosure, a multi-axis mixing apparatus is disclosed. According to the first embodiment, the apparatus includes a housing and a controller operatively connected to a first motor and a second motor. The apparatus also includes a frame supporting the first motor configured to selectively rotate a rotatable assembly. Also according to the first embodiment, the rotatable assembly supports and provides an operative connection to the second motor configured to selectively rotate a rotatable sub-assembly. Also according to the first embodiment, the rotatable sub-assembly includes a mechanism configured to receive and hold a container containing contents to be mixed, and the controller is configured to independently operate the first and second motors to mix the contents. Also disclosed is a method of mixing according to the first embodiment. Yet further disclosed is a paint mixer including the apparatus of the first embodiment, where the container is a paint container and the contents include paint to be mixed.
(7) According to a second embodiment of the present disclosure, a multi-axis mixing apparatus is disclosed. According to the second embodiment, the apparatus includes a housing and a controller operatively connected to at least a first motor configured to selectively rotate a rotatable assembly. Also according to the second embodiment, the rotatable assembly supports a rotatable sub-assembly. Also according to the second embodiment, the rotatable sub-assembly includes a mechanism configured to receive and hold a container containing contents to be mixed, and an extractor mechanism is configured to displace the container relative to the sub-assembly for removal of the container.
(8) According to a third embodiment of the present disclosure, a multi-axis mixing apparatus is disclosed. According to the third embodiment, the apparatus includes a housing and a controller operatively connected to at least a first motor configured to selectively rotate a rotatable assembly. Also according to the third embodiment, the rotatable assembly supports a rotatable sub-assembly. Also according to the third embodiment, the rotatable sub-assembly includes a mechanism configured to receive and hold a container containing contents to be mixed, and the sub-assembly includes at least one magnet configured to attract and hold a handle of the container.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
(1) FIG. **1** is an isometric view of a multi-axis mixing apparatus with a door in a closed position, according to various embodiments.
(2) FIG. **2** is a front view of the multi-axis mixing apparatus of FIG. **1**, according to various embodiments.
(3) FIG. **3** is a side view of the multi-axis mixing apparatus of FIG. **1**, according to various embodiments.
(4) FIG. **4** is an isometric view of the multi-axis mixing apparatus of FIG. **1** with the door in an open position, according to various embodiments.
(5) FIG. **5** is a front view of the multi-axis mixing apparatus of FIG. **1** with the door in the open position, according to various embodiments.

(6) FIG. **6** is a cross-sectional side view of the multi-axis mixing apparatus of FIG. **1** in the open position, according to various embodiments.
(7) FIG. **7** is a cross-sectional side view of operative mixing features of the multi-axis mixing apparatus of FIG. **1**, according to various embodiments.
(8) FIG. **8** is a top view of the operative mixing features of the multi-axis mixing apparatus of FIG. **1**, according to various embodiments.
(9) FIG. **9** is a rear view of the operative mixing features of the multi-axis mixing apparatus of FIG. **1**, according to various embodiments.
(10) FIG. **10** is an isometric view of a rotatable tumble assembly of the multi-axis mixing apparatus of FIG. **1**, according to various embodiments.
(11) FIG. **11** is a side view of the rotatable tumble assembly of the multi-axis mixing apparatus of FIG. **1**, according to various embodiments.
(12) FIG. **12** is an isometric view of the rotatable tumble assembly of the multi-axis mixing apparatus of FIG. **1** shown clamping a medium container, according to various embodiments.
(13) FIG. **13** is an isometric view of the rotatable tumble assembly of the multi-axis mixing apparatus of FIG. **1** shown clamping a large container, according to various embodiments.
(14) FIG. **14** is an isometric view of extractor components of the multi-axis mixing apparatus of FIG. **1** shown in a retracted position, according to various embodiments.
(15) FIG. **15** is an isometric view of the extractor components of the multi-axis mixing apparatus of FIG. **1** shown in an extracting position, according to various embodiments.
(16) FIG. **16** is a side view of selected extractor components of the multi-axis mixing apparatus of FIG. **1** shown with a hook in a retracted position, according to various embodiments.
(17) FIG. **17** is a side view of the selected extractor components of the multi-axis mixing apparatus of FIG. **1** shown with a hook in an intermediate extracting position, according to various embodiments.
(18) FIG. **18** is a side view of the selected extractor components of the multi-axis mixing apparatus of FIG. **1** shown with a hook in an extracting position, according to various embodiments.

DETAILED DESCRIPTION

(19) Disclosed is an easy-to-use, multi-axis mixing apparatus and related methods that use multi-axis, gyroscopic motion to mix contents in containers ranging from roughly a quart or less to larger than five U.S. gallons. Disclosed embodiments also utilize improved operative connections, simpler and improved mechanical operation, and more reliable operation. Disclosed embodiments yet further provide for improved and assisted extraction of a container from the mixing apparatus and other improvements to container handling.
(20) With reference to FIGS. **1-6** a multi-axis mixing apparatus **10** (or mixer) is shown. The mixing apparatus **10** has a housing **18** supported by a support structure **22** that preferably includes at least one leg **24**, each of which can be adjustable. A movable door **12** is provided to selectively close the housing **18** when in a lowered, closed position (e.g., as in FIGS. **1-3**), and to open the housing **18** when in a raised, open position (e.g., as shown in FIGS. **4-6**). The door **12** is preferably hinged or slidable, and is provided with a handle **14** that can be grasped by a user to move the door such that it is opened or closed when desired. As shown, a container shelf **16** is attached to the housing **18** to facilitate loading/unloading of a container (e.g., a small container, a medium container **94**, or a large container **96**) into the mixing apparatus **10**, and a control panel **20** is preferably provided on the housing **18**. The control panel **20** can include various controls, such as an emergency off feature **21** and/or various buttons and/or screens (not shown) such that a user can input various parameters for mixing using the mixing apparatus **10**. Unloading and extracting can be understood to be used interchangeably in this disclosure.
(21) The contents of a container can be liquid, and one example is paint components to be mixed within a sealed container. As used herein, a container can include any size vessel, bucket, paint can, or any other suitable container for any suitable contents. The container can include a handle **98**

attached to the various sized containers via a handle mount **100**, as described below. The handle **98** can be a bail bar, or any other suitable type of handle. In other embodiments, the handle **98** can be omitted.

(22) As shown in FIGS. **4-6**, supported within and by the housing **18** is a mounting frame **30** for supporting operative mixing features **28**, which can include various frames, motors, mechanical extraction features, and the like.

(23) The operative mixing features **28** include an upper plate **40** and a lower plate **50** that are adjustably positionable relative to each other, and can be used to securely clamp a container in place for mixing. Preferably, the upper plate **40** and lower plate are substantially parallel to each other at various and/or all positions. Generally speaking, components of the mixing apparatus **10** contained within the housing **18** are the operative mixing features **28**, although not every feature is necessarily directly or indirectly used for mixing. A cross-sectional side view of the mixing apparatus **10** is shown in FIG. **6**, which provides additional detail of certain operative mixing features **28** within the housing **18** of the mixing apparatus **10**.

(24) FIG. **7** shows the operative mixing features **28** of the mixing apparatus in greater detail. The operative mixing features **28** provide for controlled rotational, mixing movement in two different axes, as shown. Mixing with mixing apparatus **10** can occur by "gyroscopic" or multi-axis motion. Multi-axis motion means that a container is rotated according to two (or more) axes. The two axes can include a first, spin axis **54** and a second, tumble axis **52**. As shown, a tumble motor **36** mounted to the mounting frame provides rotation about the tumble axis **52** and a spin motor **38** (indirectly mounted to the mounting frame) provides rotation about the spin axis **54**. The tumble motor **36** can be a direct drive, electric motor operatively connected to a controller (not shown). Example controllers can include a hardware processor operatively coupled to at least a memory.

(25) Also with reference to FIG. **7**, the operative mixing features **28** can be variously characterized as being comprised within a rotatable tumble assembly **32** attached to tumble drive components **34**. More specifically, the tumble motor **36** of the tumble drive components **34** is fixedly attached on one end to the mounting frame **30** via a stationary tumble motor frame **26**, and attached on another end to a tumble frame **42** of the rotatable tumble assembly **32** via an attachment flange **68**. The tumble motor **36** can rotate the tumble frame **42** of the rotatable tumble assembly **32** rotate according to the tumble axis **52**. The spin axis **54** and tumble axis **52** can be perpendicular to each other. Preferably, the spin axis **54** is perpendicular to the lower plate **50** of the mixing apparatus **10**. The orientation and positioning of the axes **52**, **54** can affect where the contents travels inside the container and the velocity and shear in the contents, affecting mixing. During mixing, it is generally preferable that a center of mass of the container and contents thereof are substantially balanced according to each axis of rotation, e.g., the tumble and spin axes **52**, **54**.

(26) Comprised within the rotatable tumble assembly **32** and rotated by the spin motor **38** is a spin sub-assembly **37**, itself comprising the spin motor **38**. The spin motor **38** can be a direct drive, electric motor operatively connected to a controller (not shown). The spin motor **38** is configured to provide independent rotation about the spin axis **54** relative to the tumble motor **36**. The spin motor **38** can be operatively connected to the controller and/or a power source, among other components, via one or more slip ring (not shown) and/or any other suitable electrical connections that allow for rotation while maintaining an operative connection. The spin motor **38** is attached to the tumble assembly **32** via a spin motor support frame **58**. The upper plate **40** and the lower plate **50** can each be generally planer and circular and centered for rotation according to the spin axis **54**. The lower plate **50** preferably comprises one or more openings and/or graduated, stepped indentations for receiving various sizes of containers, as described in greater detail below.

(27) According to embodiments, separate motors **36** and **38** are provided, one for each axis **52**, **54** of rotation within the mixing apparatus **10**. Preferably, no gears or drive belts are used to effect rotation according to the spin and tumble axes **52**, **54**. For example, direct drive electric motors preferably provide simple construction and wiring. Direct drive electric motors preferably eliminate

the need for gears or belts in order to impart rotation of the various components. Direct drive motors can also reduce noise and wear during motor operation. Furthermore, direct drive motors can provide an improved and simpler ability to independently control rotational speed (RPM) of both the spin and tumble movements. The tumble **36** and/or spin motors **38** as used herein can be alternating current (AC) or direct current (DC) electric motors, among any other type of suitable motor. The motors preferably operate at between about 12 and 64 Volts, AC or DC, and preferably are rated for between 100 and 2,000 Watts, each. The motors **36**, **38** can be operated at substantially the same speed, different speeds, or at various changing speeds and/or power levels during a mixing process, among various other combinations. Furthermore, the controller (not shown) can be configured to selectively control at least one of the motors **36**, **38** to be selectively operated as a brake mechanism, e.g., short various electrical connections, or operate as a generator or the like, if and when it is desirable to slow or stop the mixing process.
(28) In preferable embodiments, at least one slip ring is utilized to provide an operative connection between a controller and at least one motor, and/or a motor and another motor within the mixing apparatus **10**. As used herein, a slip ring has at least one sliding electrical contact that rotates with a moving structure or contact relative to another structure or contact. Contemplated wiring arrangements herein also account for the upper and lower plate **40**, **50** moving toward and away from each other, and provide flexible and/or movable wiring such that an operative connection is maintained during movement of various parts of the mixing apparatus **10** before, during, and/or after mixing operation. Typical arrangements that utilize gears and/or belts typically mean that the ratio of spin and tumble are fixed and cannot be changed. By employing separate tumble and spin motors **36**, **38**, embodiments of the present disclosure allow for change in ratios of the two motors any time during a mixing process.
(29) Various container clamping features can provide linear movement of the upper plate **40** relative to the lower plate **50** such that a container can be held securely for mixing using the mixing apparatus **10**. The rotatable tumble assembly **32** preferably comprises the clamping features, which together rotate according to the tumble axis **52** during mixing operation. A linear clamping mechanism **65** is preferably generally symmetrical and can be spin balanced with or without a container clamped thereby, and comprises a clamp screw **66**, held to the tumble frame **42** by a clamp screw mount **67**. The clamp screw **66** is operatively rotatable by a clamp screw motor (see, e.g., **73** of FIG. **12**) such that one or more carriers are moved during operation of the linear clamping mechanism **65**. The container can be held by the linear clamping mechanism **65** by the two plates **40** and **50** as they move and clamp on the container containing contents to be mixed at the start of a mixing cycle.
(30) The upper plate **40**, spin motor **38**, spin motor support frame **58**, and various other upper components can be movably attached to an upper carrier **60** that is linearly moveable according to the tumble frame **42**, which can comprise one or more rails, by actuation of the clamp screw **66**. Likewise, the lower plate **50**, and various extraction components comprised within the rotatable tumble assembly **32** can also be movably attached to a lower carrier **62** that can be similar to the upper carrier **60**. The lower carrier can also be linearly movable according to the tumble frame **42** by actuation of the clamp screw **66**, preferably in a direction opposite a linear movement of the upper carrier **60**.
(31) According to various embodiments, the mixing apparatus **10** described herein can automatically and securely hold a container of various sizes when a user input is received. The user input can be received by depressing a button or turning a knob, among various other controls of the control panel **20** of the mixing apparatus **10**. As shown, the control panel **20** of the mixing apparatus **10** optionally includes the emergency off feature **21**, which can be located in any location of the mixing apparatus **10**. The user input can include a selection of a mixing cycle, speed of rotation, etc. Based on the user selection and/or automatic or detected settings, the mixing apparatus **10** preferably mixes the container for a predetermined time and according to a

predetermined mixing cycle, such as including a mixing time and/or mixing programming. The mixing time and/or programming are preferably selected according to predetermined stored settings. In some embodiments, the user can adjust an automatically set time and/or mixing cycle using the control panel **20**.

(32) Prior to and during mixing, the container can be securely held for mixing by the linear clamping mechanism **65**. More specifically, upon a determination that the mixer door **12** is closed, the upper **40** and lower plate **50** will move linearly toward each other, and clamp and securely hold the container to be mixed. The user can place the container to be held and mixed within the mixing apparatus **10** prior to the cycle. Alternatively an automated machine, which can be programmed to operate similarly to a human user, can be configured to automatically place the container in the mixing apparatus **10** before the container is also clamped automatically.

(33) The container is preferably inserted into the mixing apparatus **10** in a certain preferred orientation. For example, it may be preferable that the container is placed in the mixing apparatus **10** in an orientation such that a handle **98** of the container is movable in a dihedral arc toward an opening in the mixing apparatus **10**, e.g., toward the position of the door **12**. In some embodiments, the container is placed in the mixing apparatus **10** such that a front label is facing out, or toward the point of entry. Preferably, before loading and/or removing the container relative to the mixing apparatus **10**, the linear clamping mechanism **65** is latched or locked in place so the lower plate **50** supporting the container is in a lowered position and the container is face or label forward. Upon a mixing process ending, the mixing apparatus **10** can return the container to an upright and face-out position, and rotate the container until the handle **98** is facing forward.

(34) Also shown in FIG. **7** are various extraction and/or loading components of the mixing apparatus **10**. It may be desirable to provide assistance to a user or other automated machine for extraction of a container after mixing, or during loading before mixing. Specifically, a rolling component such as a loading roller **48** is shown adjacent to the lower plate **50**, at a point where a container would be initially placed in the mixing apparatus **10** to assist in loading or unloading a heavy container. Also shown is a pusher mechanism that includes a pusher frame **44** with a pusher tab **64** attached to a crossbar portion of the pusher frame **44** such that a container can be contacted and pushed by the pusher tab **64** when the pusher frame is moved. Yet further shown is another rolling component, a vertical extraction roller **46**, that is moveable such that it penetrates an opening (see **75** of FIG. **12**) in the lower plate **50** when desired for extraction of a container. In some embodiments, the front panel of the machine comes off to make the machine ready and/or accessible for automation. As used herein, a rolling component is a general term that can denote a roller, a roller bearing, or any other type of rolling or bearing configuration.

(35) With reference now in particular to FIGS. **8-11**, various specific parts of the operative mixing features **28** of the mixing apparatus **10** are shown. As shown in FIG. **8**, on the upper plate **40** of the spin sub-assembly **37** are one or more fasteners **71**, which can be used to secure various components of the spin sub-assembly together during assembly. With reference now to FIG. **9**, the upper plate also comprises one or more magnets **70** positioned around a perimeter of the upper plate **40**. The magnets **70** can be any suitable type of magnet, (e.g., permanent or electromagnet) and are attached to the upper plate **40** directly or indirectly. The magnets **70** are positioned to attract and secure a handle **98** (see, e.g., FIG. **13**), such as a metal handle **98** of a container during mixing. A flexible connection **90** is also shown that preferably provides an operative connection to various components of the rotatable sub-assembly **32** and allow for linear movement of the various carriers **60**, **62** while maintaining the operative connection. As shown, the flexible connection **90** is a semi-flexible chain-style connection that provides a controlled bending of various cables and/or connection without unduly pinching or stressing the various connections.

(36) In previously existing configurations, an elastomeric cord or a spring with a hook on an end were manually attachable to a bail or handle of a container for holding during mixing. The cord or spring was for example located a rear portion of the mixer interior. These manual arrangements

have had drawbacks, such as generally requiring that the handle be at the rear of the container. This made it inconvenient to grasp the handle when inserting and removing the container. As the existing configurations were manual, extra time was required for attachment and it was also possible for a user to forget to attach the bail or handle of the container to the cords or spring such that unwanted bail or handle movement during mixing were avoided.

(37) In this disclosure, it is contemplated that the container is inserted into the mixing apparatus **10** with the handle **98** facing forward or outward. Once the container is inserted into the mixing apparatus **10**, and as the container approaches the upper plate **40**, the magnets **70** automatically attract and hold the handle **98** before mixing. In some examples, the magnets **70** attract and hold the handle **98** after the container is clamped by the linear clamping mechanism **65**. Holding the handle **98** during mixing can reduce undesirable noise since the handle **98** would not move relative to the container, and would not strike other components as the container is moved and mixed.

(38) The magnets **70** can provide a handle **98** holding feature for various container sizes and configurations. By automatically holding the handle **98** during mixing, the effort and time expended by a user can be reduced, and there is no longer a need to apply a hook to hold the handle **98** in place. The bail or handle **98**-holding function also improves the user experience by reducing or eliminating the likelihood that a user disadvantageously neglects to secure the handle **98** prior to mixing. The handle holding magnet **70** also assists automation since the handle **98** position is predictable and controlled. The magnets **70** automatically let go of the handle **98** when the linear clamping mechanism **65** is opened, thus pulling the handle **98** out of the effective reach of magnets **70**.

(39) Carriers **60**, **62** provide smooth, controlled movement of the plates **40** and **50** of the linear clamping mechanism **65**, among other components of the rotatable tumble assembly **32**. Each carrier **60**, **62** comprises multiple carrier wheels **76** that are configured to roll along the tumble frame **42** in a controlled, linear manner when the linear clamping mechanism **65** is operated. Preferably, and as shown, the carriers **60**, **62** are provided with carrier wheels **76** on at least two sides of the tumble frame **42**, and preferably on three or more sides of the tumble frame in order to guide the carriers **60**, **62** predictably and smoothly. Also as shown, the carrier wheels **76** are provided in tandem pairs, such that stability is yet further improved. The carrier wheels **76** can utilize roller bearings in order to provide smooth, consistent, and long-lasting performance. Preferably, rolling components such as sealed roller bearings are used.

(40) In preferable embodiments, the mixing apparatus **10** can be entirely or substantially free of grease for lubrication of various mechanisms. For example, the upper and lower carriers **60**, **62** can move on a set of rails of the tumble frame **42** using rolling components such as bearings within wheels **76**, such as roller bearings as described herein. In existing mixers, it is typical to use bronze against steel, which typically requires greasing. Friction of a roller bearing is generally lower than a corresponding arrangement using greased bronze. Clamp force is also generally more constant over time when greasing of various parts is avoided, as grease can spread, dissipate, or otherwise become less functional over time. For example, instead of grease, examples of carriers (such as **60**, **62**) and/or nuts (e.g., extraction hook carrier **106**) that rides on various screws (e.g., leadscrews) can be made of a lubricated engineered plastic, preferably avoid periodic maintenance and lubrication.

(41) Various sized and shaped indentations in the lower plate **50** configured to receive various sized containers for mixing. The various indentations are shown best at FIGS. **10**, **14**, and **15**, and can provide a tiered, stepped-type arrangement. As shown, a small indentation **82** for receiving a small container (not shown) is lowest positioned and narrowest in diameter, a medium indentation **84** for receiving a medium container **94** is positioned higher (shallower) and wider than the small indentation, and a large indentation **86** is positioned higher (yet shallower) and widest for receiving a large container **96**. Although certain shapes and sizes of indentations are shown, different, fewer, additional, or any other configuration of the lower plate **50** is also contemplated. The various

indentations can instead take the form of posts, notches, or any other protrusions without departing from the scope of this disclosure.

(42) With reference now in particular to FIGS. **12** and **13**, various sizes of containers are shown as they are clamped by the upper and lower plates **40**, **50** of the spin sub-assembly **37**. Specifically, FIG. **12** shows the medium container **94** located on the lower plate **50** and fitted into the medium indentation **84**. FIG. **13** shows the large container **96** located on the lower plate **50** and fitted into the large indentation **86**.

(43) The medium container **94** as shown in FIG. **12** has a handle mount **100**, although no bail or handle is shown. In FIG. **13**, a handle **98** is shown mounted to the handle mount **100** of the large container **96**. Also as shown, the handle **98** (or handle) is movable along a dihedral angle as it is rotatably mounted at two ends to bail mounts **100**. Also as shown in FIG. **13**, the handle **98** can be composed of a metal, and more specifically of a ferromagnetic metal, that can be attracted to one or more bail magnets **70** of the upper plate **40** when the container (e.g., large container **96**) is placed on the lower plate **50** and clamped by the linear clamping mechanism **65**.

(44) FIGS. **14** and **15** show isometric views of an example extractor mechanism **102** of the mixing apparatus **10** at various stages in an extraction procedure. More specifically, FIG. **14** shows a retractable extractor mechanism **102** in a retracted position, such as it would be during a mixing process. FIG. **15** shows the extractor mechanism **102** in an extracting position as a container would be extracted, e.g., after mixing. In other embodiments, the extractor mechanism utilizes an extractor arm that does not pivot or retract for operation, e.g., a “fixed,” or non-pivoting style extractor arm (not shown). It is to be understood that the shown embodiment is merely one possible example of an extractor mechanism and that a different types of extractor and extractor arm configurations are also contemplated within the present disclosure.

(45) The extractor mechanism **102** is preferably configured to displace a container relative to the lower plate **50** of the spin sub-assembly **37**. FIG. **14** shows the extractor mechanism **102** in a retracted position, and FIG. **15** shows the extractor mechanism **102** in an extracting position. As shown in FIG. **15**, the extractor mechanism **102** comprises a lifting or lever extractor arm **74** attached at arm connection **89** and that pivots at bolt **108** such that extraction roller **46** selectively passes through the opening **75** in the lower plate **50**. As shown, the extractor mechanism **102** also comprises a pan **78** positioned below the lower plate **50**. The pan **78** comprises a linear slot **79** configured to permit an extractor hook **80** to selectively penetrate the pan **78** when the hook **80** is moved during container extraction. Operatively connected to the pusher frame **44** is a hook catch bar **81**. As the hook **80** is moved during extraction, the hook **80** contacts and pushes on the hook catch bar **81**, thus moving the pusher frame **44** and the container to be extracted. In some embodiments, the pan **78** under the rotating frame is removable, thus facilitating easy cleaning of any spilled, leaked, or otherwise collected contents. Extractor mechanism, as shown in FIGS. **14** and **15**, also includes the opening **75**, the extraction roller **46**, various indentations (e.g., small indentation **82**), and the pusher tab **64**.

(46) Containers, and in particular a large container **96**, can be beneficially extractable with the assistance of the extractor mechanism **102**. Ledges in the lower plate **50**, such as of the various indentations, may catch on various portions of containers of varying sizes. In various embodiments, the shape of a supporting lower plate **50** can allow for automatic or assisted extraction of containers. Large and/or heavy containers benefit particularly from assisted extraction after mixing.

(47) Shown best with reference to FIGS. **16-18**, the hook **80** is operatively connected to an extractor motor unit **88** that moves the hook using a linear actuator, such as a rotatable extractor screw **104** that when rotated by the extractor motor unit **88** causes an extraction hook carrier **106** attached to the hook **80** to move linearly during extraction. The extractor motor unit **88** output shaft **126** is operatively connected to the extractor screw **104** by a drive coupling **110** as shown. The extractor screw **104** can be rotatably held in place by a stationary motor side mount **116** and a stationary distal mount **114**, as shown.

(48) In some embodiments, a user or an external automated mechanism (not shown) can insert and/or remove the container from the mixing apparatus **10** before or after mixing. The automated mechanism would potentially have difficulty properly positioning the handle **98**. The extractor mechanism **102** described herein can assist such automated mechanism by automatically removing a container from the mixing apparatus **10**, such as with a press of a button. The loading and/or extraction rollers **48**, **46** under the front edge of the container lifts and/or supports the container. This lifts and/or facilitates movement of the container over any ledge that may be present. A ramp can also be provided for roller vertical movement. The combination of extractor mechanism **102** features therefore makes manual and/or automatic loading or unloading of containers easier than before.

(49) FIGS. **16-18** show various example steps of the extractor mechanism **102** as it operates. FIG. **16** shows a first step in an extraction process of a container for use with the mixing apparatus **10**. As shown, FIG. **16** shows an embodiment where an optional flip up hook **80** is used, that retracts when not being operatively used. FIG. **16** shows the extraction mechanism in a default, retracted position, and FIGS. **17** and **18** show sequential extraction positions as the container is extracted and where the hook **80** extends through the pan **78** for use in extraction. FIG. **17** shows an intermediate extraction position, and FIG. **18** shows a more fully extracted position of the extraction mechanism **102**.

(50) In preferable embodiments, various sensor components determine a position of the hook carrier **106**. For example, it may be desirable to assure that the hook carrier **106** and hook **80** are retracted before starting a mixing process. As shown best in FIG. **17**, a magnet **111** is coupled to the hook carrier **106** and a magnet sensor **112** is coupled to the motor side mount **116**. When the magnet **111** is proximate the sensor **112** a signal can be sent to a controller to indicate that the hook **80** and carrier **106** are retracted. Other suitable forms and examples of sensors and arrangements are also contemplated herein.

(51) The hook **80** preferably is either always vertically oriented (e.g., fixed embodiments) or the hook springs cause the hook **80** to rotate up from a more horizontal position to a more vertical, latching position to push the hook catch bar **81** during operation. The pusher tab **64** of the pusher frame **44** slides the container forward on the lower plate **50** and roller(s) **46**, and/or **48** until the container is easily removed. In embodiment where the pusher frame **44** is pulled by a flip up hook **80**, one or more biasing element such as a spring **24** can retract the hook **80** when not being used to extract the container. The spring **24** can be any form of biasing element, such as a leaf spring, clock spring, coil spring, etc. A bearing ramp **122** can protrude downward, thus applying pressure on a hook bearing **120** of the hook **80** when the hook **80** is retracted. The shape of the bearing ramp **122** can provide a certain amount of hook **80** movement and/or rotation as the hook **80** is retracted.

(52) In various embodiments, including where the hook **80** is pivotable such that it flips or rotates up or down or is fixed in a vertical orientation, the linear (e.g., horizontal) position of the hook **80** can be motivated by the extractor motor unit **88**. In the example when the hook **80** is pivotable, a hook pivot bolt **118** can provide a pivot axis to the hook **80**. The hook spring **124** can provide a bias to the hook **80**, for example, such that it is in a flipped up position by default unless a contact between a hook bearing **120** and a bearing ramp **122** causes the hook to be flipped down when retracted.

(53) According to some embodiments, at a lower portion of the mixing apparatus **10**, the extractor motor unit **88** rotates the extractor screw **104** that pushes the hook **80** attached to the extraction hook carrier **106** forward. In other embodiments, a belt or other drive or linear actuator can be used to move the hook **80**.

(54) As used herein, the singular forms “a,” “an,” and “the” encompass embodiments having plural referents, unless the content clearly dictates otherwise. As used in this specification and the appended claims, the term “or” is generally employed in its sense including “and/or” unless the content clearly dictates otherwise.

(55) Unless otherwise indicated, all numbers expressing feature sizes, amounts, time periods, and physical properties are to be understood as being modified by the term “about.” Accordingly, unless indicated to the contrary, the numerical parameters set forth are approximations that can vary depending upon the desired properties sought to be obtained by those skilled in the art utilizing the teachings disclosed herein.
(56) The present invention has now been described with reference to several embodiments thereof. The entire disclosure of any patent or patent application identified herein is hereby incorporated by reference. The detailed description and examples have been given for clarity of understanding only. No unnecessary limitations are to be understood therefrom. It will be apparent to those skilled in the art that many changes can be made in the embodiments described without departing from the scope of the invention. Thus, the scope of the present invention should not be limited to the structures described herein, but only by the structures described by the language of the claims and the equivalents of those structures.
(57) Selected examples of the present disclosure are provided below:
(58) A first example includes a multi-axis mixing apparatus, comprising: a housing; a controller operatively connected to a first motor and a second motor; a frame supporting the first motor configured to selectively rotate a rotatable assembly; and the rotatable assembly supporting and providing an operative connection to the second motor configured to selectively rotate a rotatable sub-assembly, wherein the rotatable sub-assembly comprises a mechanism configured to receive and hold a container containing contents to be mixed, and wherein the controller is configured to independently operate the first and second motors to mix the contents.
(59) According to some embodiments of the first example, the first motor causes a rotation about a first axis, and the second motor causes a rotation about a second axis. According to some embodiments of the first example, the first axis is perpendicular to the second axis. According to some embodiments of the first example, at least one of the first motor and second motor is a direct drive motor. According to some embodiments of the first example, the mechanism of the rotatable sub-assembly is a linear clamping mechanism. According to some embodiments of the first example, the linear clamping mechanism comprises a first plate that is adjustably positionable relative to a second plate that is substantially parallel to the first plate at various positions. According to some embodiments of the first example, at least one of the first plate and the second plate has at least one indentation for receiving a portion of the container.
(60) According to some embodiments of the first example, the apparatus further comprises an extractor mechanism configured to displace the container relative to the sub-assembly. According to some embodiments of the first example, the sub-assembly comprises an opening through which a lifting extractor arm of the extractor mechanism can pass. According to some embodiments of the first example, the lifting extractor arm is fully withdrawn from the opening when the apparatus is in use. According to some embodiments of the first example, the extractor mechanism comprises a linear actuator and an extractor hook that is caused to be selectively extended and rotated by movement of the linear actuator such that the extractor hook contacts the container indirectly or directly upon operation of the linear actuator, and selectively displaces the container from the sub assembly. According to some embodiments of the first example, the linear actuator comprises a leadscrew mechanism.
(61) According to some embodiments of the first example, at least one of the first and second plates comprises at least one magnet configured to attract and hold a handle of the container. According to some embodiments of the first example, the handle is composed of metal. According to some embodiments of the first example, the at least one magnet is configured to attract the handle of the container when the apparatus is in use. According to some embodiments of the first example, the container is a paint container and the contents include paint to be mixed. According to some embodiments of the first example, the apparatus further comprises a power supply unit operatively connected to at least the first motor and the second motor. According to some embodiments of the

first example, at least one connection comprises a slip ring connection. According to some embodiments of the first example, the controller selectively operates the first motor at a different speed than the second motor during a mixing process. According to some embodiments of the first example, the controller selectively operates the first motor at substantially the same speed as the second motor during a mixing process. According to some embodiments of the first example, the controller selectively operates the first motor and the second motor at various changing speeds and/or power levels during a mixing process.

(62) According to some embodiments of the first example, the first motor is a tumble motor and the second motor is a spin motor. According to some embodiments of the first example, at least one of the first and second plates of the sub-assembly comprises a rolling component configured to operatively support the container during a container extraction process. According to some embodiments of the first example, the controller is configured to selectively control at least one of the first motor and the second motor is to selectively operate the first motor and/or the second motor as a brake mechanism. According to some embodiments of the first example, the apparatus is substantially free of grease for lubrication of various mechanisms. According to some embodiments of the first example, the apparatus comprises lubricated engineered plastic for lubrication of at least one component. According to some embodiments of the first example, the apparatus comprises at least one roller bearing.

(63) Also disclosed is a method of mixing according to the operation of the apparatus of the first example. Also further contemplated is a paint mixer, comprising the apparatus of the first example, where the container is a paint container and the contents include paint to be mixed.

(64) A second example includes a multi-axis mixing apparatus, comprising: a housing; a controller operatively connected to at least a first motor configured to selectively rotate a rotatable assembly; and the rotatable assembly supporting a rotatable sub-assembly, wherein the rotatable sub-assembly comprises a mechanism configured to receive and hold a container containing contents to be mixed, and an extractor mechanism is configured to displace the container relative to the sub-assembly for removal of the container.

(65) A third example includes a multi-axis mixing apparatus, comprising: a housing; a controller operatively connected to at least a first motor configured to selectively rotate a rotatable assembly; and the rotatable assembly supporting a rotatable sub-assembly, wherein the rotatable sub-assembly comprises a mechanism configured to receive and hold a container containing contents to be mixed, and wherein the sub-assembly comprises at least one magnet configured to attract and hold a handle of the container.

(66) A fourth example includes a multi-axis mixing apparatus, comprising: a housing; a controller operatively connected to at least a first motor configured to selectively rotate a rotatable assembly; and the rotatable assembly supporting a rotatable sub-assembly, wherein the rotatable sub-assembly comprises a mechanism configured to receive and hold a container containing contents to be mixed, the mechanism comprising at least one roller track and at least two sealed bearings connector to a carrier, the carrier configured to move linearly along the roller track such that the mechanism operates to move linearly to clamp the container for mixing.

Claims

1. A multi-axis mixing apparatus, comprising: a housing; a rotatable assembly and a rotatable sub-assembly; a controller operatively connected to a first motor and a second motor; a frame supporting the first motor configured to selectively rotate the rotatable assembly; and the rotatable assembly supporting and providing an operative connection to the second motor configured to selectively rotate the rotatable sub-assembly, wherein the rotatable sub-assembly comprises a linear clamping mechanism comprising a first plate, a second plate, and a third motor configured to adjustably position the first plate relative to the second plate, wherein at least one of the first and

second plates of the linear clamping mechanism is configured to move linearly to clamp and hold a sealed container containing contents to be mixed, wherein the controller is configured to independently operate the first and second motors to mix the contents, and wherein both the first motor and second motor are direct drive electric motors.
2. The apparatus of claim 1, wherein the first motor causes a rotation about a first axis, and the second motor causes a rotation about a second axis that is perpendicular to the first axis.
3. The apparatus of claim 1, wherein the second plate is substantially parallel to the first plate at various positions.
4. The apparatus of claim 1, wherein at least one of the first plate and the second plate has at least one indentation that is shaped and sized for receiving a portion of the container.
5. The apparatus of claim 1, wherein at least one of the first and second plates comprises at least one magnet feature configured to attract and hold a handle of the container, wherein the handle is composed of metal, and wherein the at least one magnet feature is configured to attract the handle of the container when the apparatus is in use.
6. The apparatus of claim 1, wherein at least one of the first and second plates of the sub-assembly comprises a rolling component configured to operatively support the container during a container extraction process.
7. The apparatus of claim 1, further comprising an extractor mechanism configured to displace the container relative to the sub-assembly.
8. The apparatus of claim 7, wherein the extractor mechanism comprises a linear actuator and an extractor hook that is caused to be selectively extended by movement of the linear actuator such that the extractor hook contacts the container indirectly or directly upon operation of the linear actuator, and selectively displaces the container from the sub-assembly, wherein the linear actuator comprises a leadscrew mechanism.
9. The apparatus of claim 1, wherein the container is a paint container and the contents include paint to be mixed.
10. The apparatus of claim 1, wherein at least one operative connection comprises a slip ring connection.
11. The apparatus of claim 1, wherein the controller selectively operates the first motor at a different speed than the second motor during a mixing process.
12. The apparatus of claim 1, wherein the controller selectively operates the first motor at substantially the same speed as the second motor during a mixing process.
13. The apparatus of claim 1, wherein the controller selectively operates the first motor and the second motor at various changing speeds and/or power levels during a mixing process.
14. The apparatus of claim 1, wherein the controller is configured to selectively control at least one of the first motor and the second motor is to selectively operate the first motor and/or the second motor as a brake mechanism.
15. The apparatus of claim 1, wherein the apparatus is substantially free of grease for lubrication of various mechanisms, and wherein the apparatus comprises lubricated engineered plastic for lubrication of at least one component.
16. The apparatus of claim 3, wherein the linear clamping mechanism comprises at least one roller bearing.
17. A multi-axis mixing apparatus, comprising: a housing; a rotatable assembly and a rotatable sub-assembly; a controller operatively connected to and configured to independently operate at least: a first direct drive electric motor configured to selectively rotate the rotatable assembly about a first axis; and a second direct drive electric motor; and the rotatable assembly supporting and providing an operative connection to the second direct drive electric motor configured to selectively rotate the rotatable sub-assembly about a second axis that is different from the first axis, wherein the rotatable sub-assembly comprises a linear clamping mechanism comprising a first plate, a second plate, and a third motor configured to adjustably position the first plate relative to the second plate, wherein at

least one of the first and second plates of the linear clamping mechanism is configured to move linearly to clamp and hold a sealed container containing contents to be mixed, and an extractor mechanism is configured to displace the container relative to the sub-assembly for removal of the container.

18. The apparatus of claim 17, wherein the linear clamping mechanism comprises at least one roller track and at least two sealed bearings connected to a carrier, the carrier configured to move linearly along the roller track such that the linear clamping mechanism operates to move linearly to clamp the container for mixing.

19. A multi-axis mixing apparatus, comprising: a housing; a rotatable assembly and a rotatable sub-assembly; a controller operatively connected to at least a first direct drive electric motor configured to selectively rotate the rotatable assembly; and the rotatable assembly supporting the rotatable sub-assembly, wherein the rotatable sub-assembly comprises a linear clamping mechanism comprising a first plate, a second plate, and a second motor configured to adjustably position the first plate relative to the second plate, wherein at least one of the first and second plates of the linear clamping mechanism is configured to move linearly to clamp and hold a sealed container containing contents to be mixed, and wherein the sub-assembly comprises at least one magnet feature that extends around a perimeter of the first plate, wherein the at least one magnet feature is configured to attract and hold a handle of the container.

20. The apparatus of claim 4, wherein at least one of the first plate and the second plate has at least two indentations in a stepped arrangement for receiving a portion of the container selected from at least two container sizes, and wherein the at least two indentations comprise a smaller indentation configured to receive a smaller container, the smaller indentation being positioned lower and being narrower in diameter, and a larger indentation being larger in diameter and configured to receive a larger container that is positioned higher and wider than the smaller indentation.